MASTER'S THESIS DEFENSE

Enhancing Stability in Direct Training of Spiking Neural Networks

Membrane Potential Initialization & Threshold-Robust Surrogate Gradients

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Why Spiking Neural Networks?

DNNs are powerful but expensive

Training and serving large models routinely consumes megawatt-scale power; even edge inference is hard under tight energy budgets.

SNNs: an event-driven alternative

Information flows as discrete spikes over time. Paired with neuromorphic hardware, computation skips inactive neurons — orders of magnitude lower energy.

But direct training is unstable

Recent direct-training methods narrowed the accuracy gap, yet two persistent instabilities still hold SNNs back.

THIS THESIS

Two instabilities, two principled fixes.

① **Temporal Covariate Shift**

in the membrane potential — invisible to existing remedies.

② **Gradient pathology**

when the firing threshold becomes a learnable parameter — never explicitly characterized before.

Spiking Neurons in 60 Seconds

Three discrete-time updates, plus one trick to get gradients through them

Leaky Integrate-and-Fire neuron

Integrate

$$\mathrm{M[t]} = (1-1/\tau)\cdot \mathrm{U[t-1]} \;+\; (1/\tau)\cdot \mathrm{I\_in[t]}$$

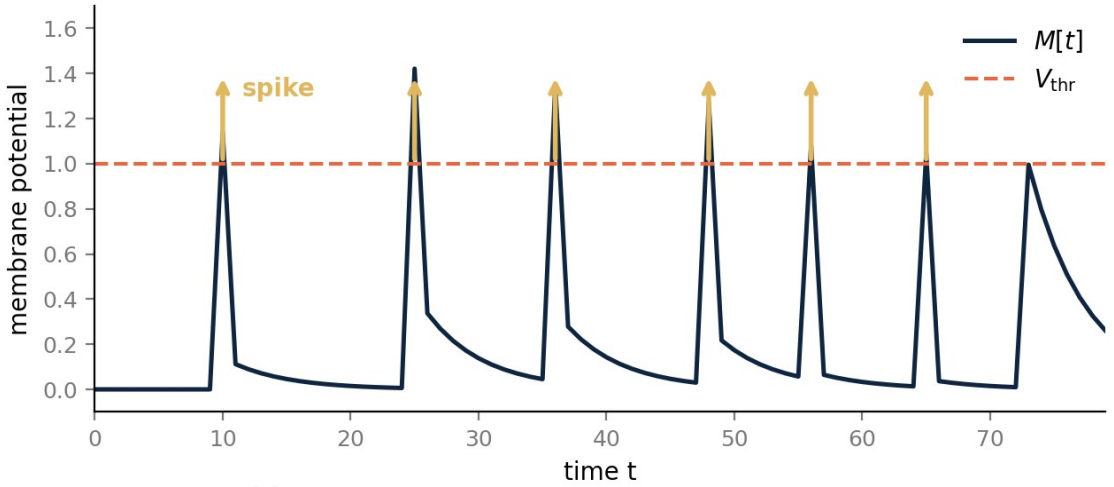
Fire

$$\mathrm{S[t]} = \mathrm{H}(\,\mathrm{M[t]} - \mathrm{V\_thr}\,) \in \{0,\,1\}$$

Reset

$$\mathrm{U[t]} = \mathrm{M[t]} - \mathrm{V\_thr}\cdot \mathrm{S[t]} \quad \text{(soft reset)}$$

Two learnable parameters: τ (leakage) · V_thr (threshold).



M[t] integrates inputs, fires at V_thr, resets — repeat.

Heaviside H is not differentiable

Surrogate gradient: backward pass uses smooth f'(x) ≈ H'(x).

Everything in this thesis turns on M[t] and V_thr · TCS at M[t] · pathology at V_thr

Surrogate Gradient — Training Through the Spike

Spikes are non-differentiable; smooth surrogates re-open backprop

The problem

$$\text{Forward:} \quad S = H(M - V\_thr) \in \{0, 1\}$$

$$\text{Backward:} \quad \partial S / \partial M = \delta(M - V\_thr)$$

Heaviside has zero gradient almost everywhere, with a delta spike exactly at threshold — gradient information cannot flow.

The fix: surrogate gradient (SG)

Forward unchanged. In backward, replace δ with smooth f'(x):

$$\partial S / \partial M \approx f'(x), \quad |x| < \gamma/2$$

f' is peak-normalized (max = 1), supported on a window of width γ around 0. Forward stays binary; gradient flows.

Common shapes · same window, different curve

▶ Rectangular ▶ Triangular ▶ Arctan ▶ Sigmoid

Shape matters less than scale — TrSG works across all four.
γ controls the active window WIDTH; height is set by the chain rule.

Two natural choices for x

x = M − V_thr → AS-SG (Absolute scale)
x = M / V_thr − 1 → RS-SG (Relative scale)

Same SG curve, two ways to plug in V_thr → analyzed in Part II.

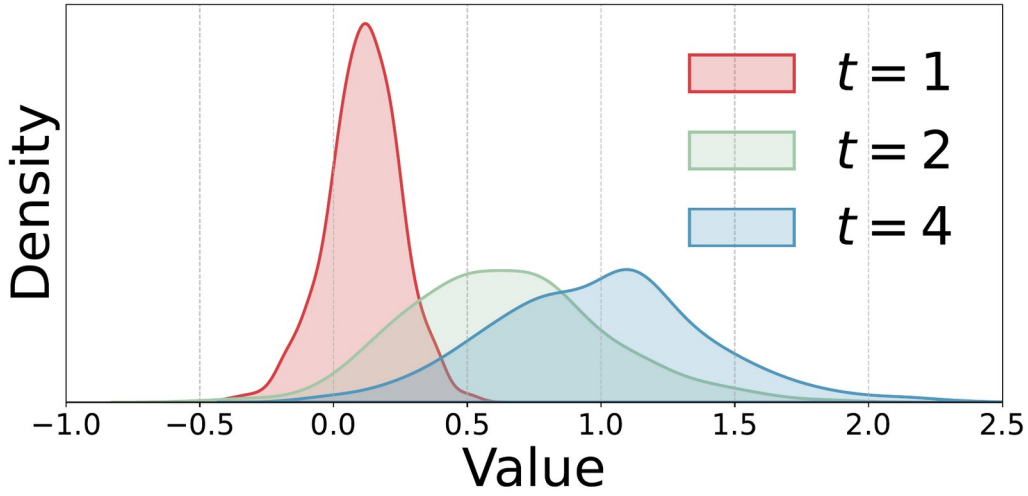
Diagnosis ①: TCS Lives in the Membrane Potential

What everyone notices

Activation distributions drift across timesteps. Batch-norm statistics estimated at one timestep no longer fit at another — accuracy drops.

What existing fixes do

TEBN, TAB add per-timestep BN parameters.
More compute, more parameters — and they don't reach the cause.



M[t] under tdBN — distribution clearly drifts as t grows

Our claim

TCS originates one level deeper — in M[t] itself.

Diagnosis ②: Trainable V_thr Breaks the Gradient

PLIF, LTMD, DIET-SNN all let V_thr learn — but the surrogate gradient is scale-sensitive

V_thr ≪ 1 (small threshold)

- AS-SG floods — fixed window covers almost the entire distribution
- RS-SG explodes — the 1/V_thr factor blows the magnitude up
- Real example: at V_thr = 0.1, RS-SG diverges to NaN within 100 epochs

V_thr ≫ 1 (large threshold)

- AS-SG starves — fixed window covers a tiny tail of the distribution
- RS-SG vanishes — the same 1/V_thr factor shrinks the magnitude
- Real example: at V_thr = 2.0, AS-SG hits 0% active neurons by epoch 100

These are not corner cases.

Trainable V_thr in practice drifts well below 1 in early layers and above 2 in deep ones — exactly the regimes where AS-SG and RS-SG both fail. We'll quantify in Part II.

Why Does the Membrane Potential Drift?

Because it's a Markov chain pulled toward a unique distribution

The recurrence is a Markov chain

U[t+1] = f(U[t], I_in[t+1]) — Markov by construction; i.i.d. inputs ⇒ time-homogeneous

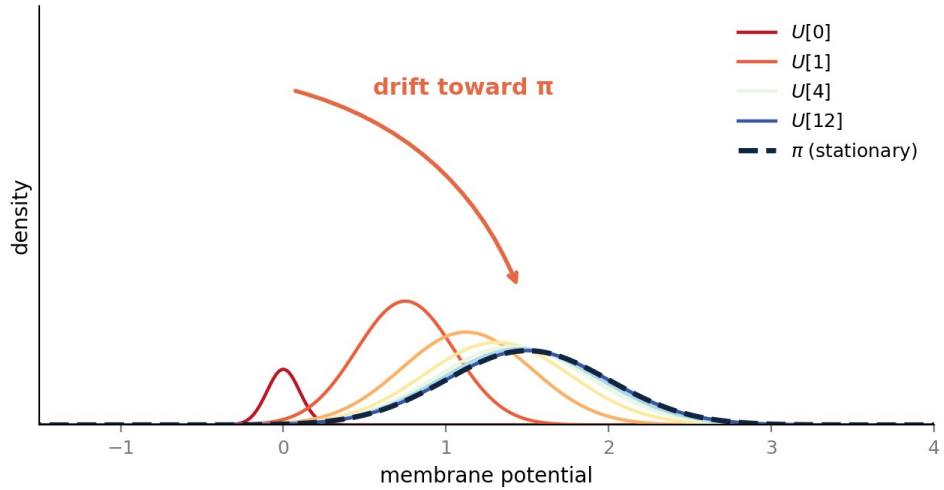
Theorem (Doeblin minorization)

Under (i.i.d. inputs) + (bounded U[t]),
the chain has a unique stationary π. Moreover,

$$\| P^n(x, \cdot) - \pi(\cdot) \|_{TV} \leq C \cdot (1 - \varepsilon)^n$$

Implications

- Initial state ≠ π ⇒ drift is inevitable
- Standard U[0] = 0 is far from π
- Don't fix BN — fix the initial state



Distribution of U[t] for t = 0, 1, 2, 3, 4, 5, 8, 12

Starting from U[0]=0 (sharp red), the distribution exponentially approaches the stationary π (dashed navy).

MP-Init — Initialize at the Stationary Mean

U[T] is the closest sample to π → use a running mean of E[U[T]]

Lemma — best constant is the mean

$$\operatorname{argmin}_c \; \mathbb{E}[\,(\pi - c)^2\,] \;=\; \mathbb{E}[\pi]$$

Practical estimator

U[T] is the closest sample to π. Maintain a per-layer running mean
μ ← β · μ + (1 − β) · mean(U[T]) over training, β = 0.9.
At every batch, set U[0] = μ.

Algorithm (training-only)

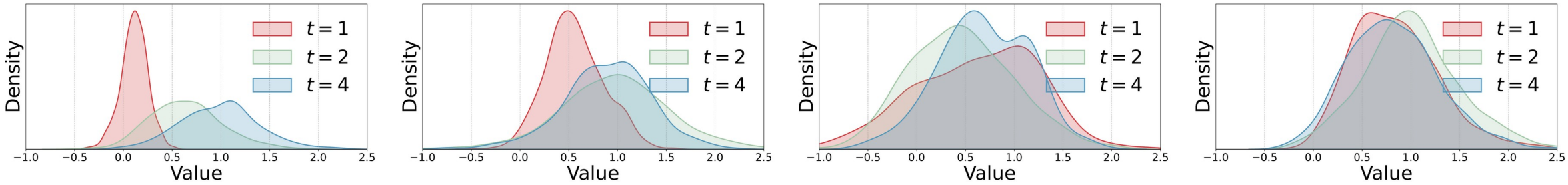
```
init  μ_l ← 0   for each layer l

for each batch:
    U_l[0] ← μ_l
    run T LIF steps as usual
    m ← mean( U_l[T] of active neurons )
    μ_l ← β · μ_l  +  (1 − β) · m
```

Inference: μ frozen → standard LIF dynamics

Zero changes to BN, no per-timestep parameters, full neuromorphic-hardware compatibility.

MP-Init Works — in Both Distribution and Metric

Membrane potential distributions across timesteps (ResNet-19 / CIFAR-100)



tdBN | TEBN | TAB | **MP-Init (ours)**

Consistent accuracy gain on every baseline

Setting	Baseline	+ MP-Init
CIFAR-100 · T=2 · tdBN	72.35	**74.01**
CIFAR-100 · T=4 · tdBN	75.55	**76.09**
CIFAR-100 · T=4 · TEBN	75.96	**76.45**
CIFAR-100 · T=4 · TAB	76.25	**77.24**
DVS-CIFAR10 · T=10 · tdBN	76.60	**77.37**

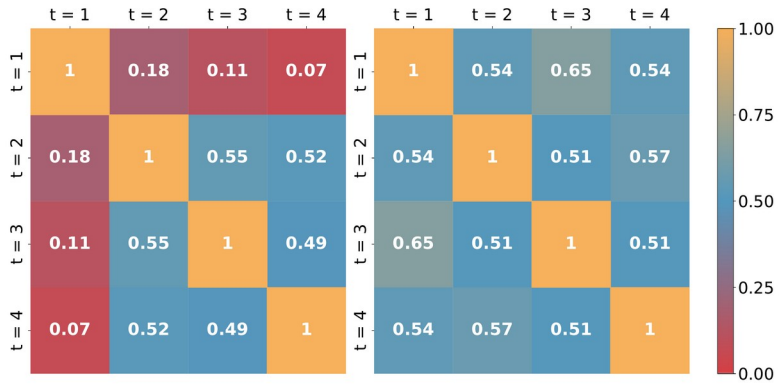
+1.66 %pt at T = 2

Biggest gain in the most latency-critical regime — exactly where SNN energy efficiency matters.

Why MP-Init Works — A Closer Look

Aligning distributions aligns gradients — and improves the firing-rate frontier

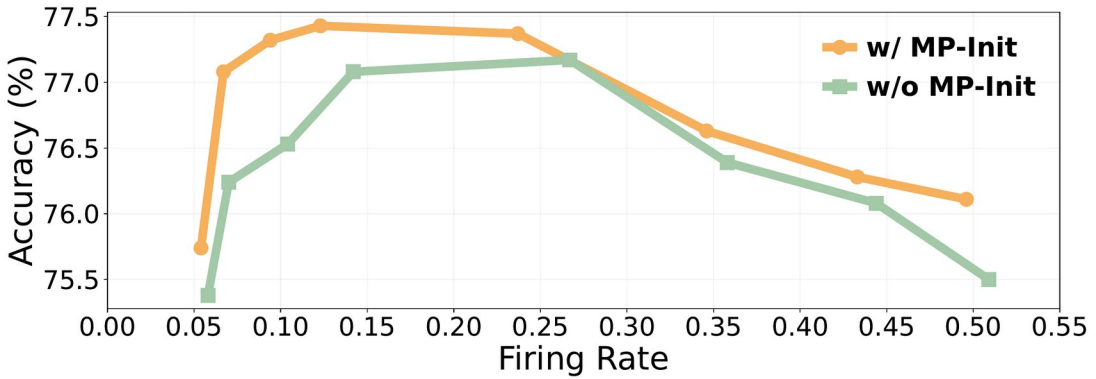
Gradient cosine similarity across timesteps



ResNet-19 · CIFAR-100 · 4 timesteps

- Without MP-Init: gradients at t=1 conflict with later steps (cos 0.07–0.18)
- With MP-Init: temporal ensemble realigned (cos ≈ 0.5)
- Stable optimization — not just stable inference

Accuracy vs. firing rate (Pareto frontier)



ResNet-19 · CIFAR-100 with sparsity regularizer

- MP-Init dominates baseline at every firing rate
- Even at 6% firing rate, beats all baseline settings
- Gains come from spiking BETTER, not spiking MORE

Higher accuracy from more stable training — efficiency-preserving improvements, not accuracy-via-spend.

Two Surrogate-Gradient Families

AS-SG and RS-SG — look similar, behave very differently when V_thr is trainable

∂S/∂M ≈ f'(x) · rectangular f' = 1 on | x | < γ/2 (peak-normalized) · γ = active window width

AS-SG · Absolute-Scale

$$x = M[t] - V\_thr$$

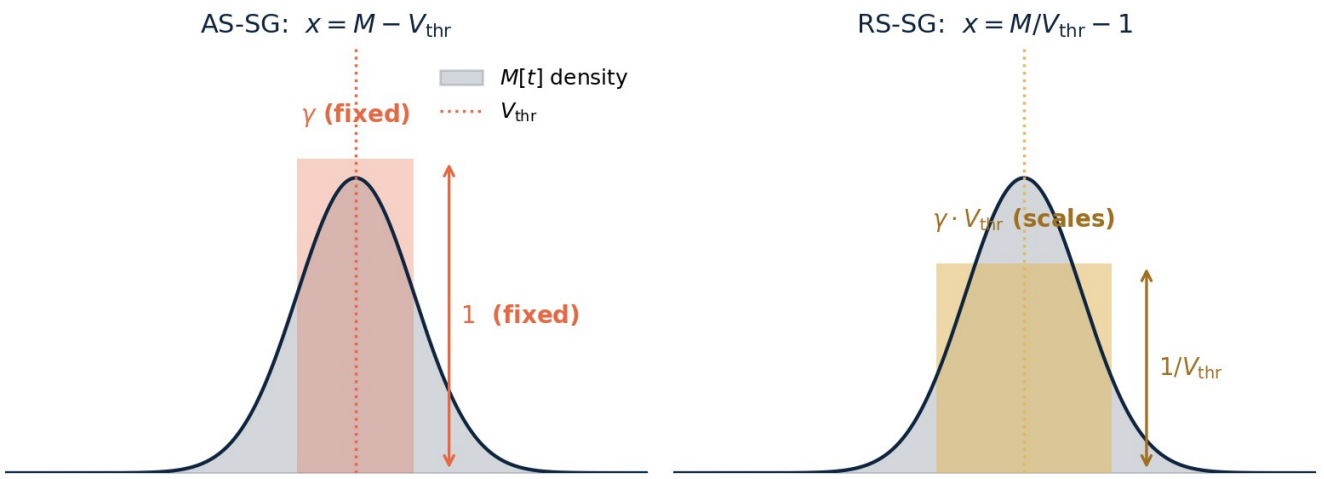
Window of FIXED width γ, centered at V_thr

RS-SG · Relative-Scale

$$x = M[t] \,/\, V\_thr - 1$$

Window SCALES with V_thr — but magnitude / V_thr

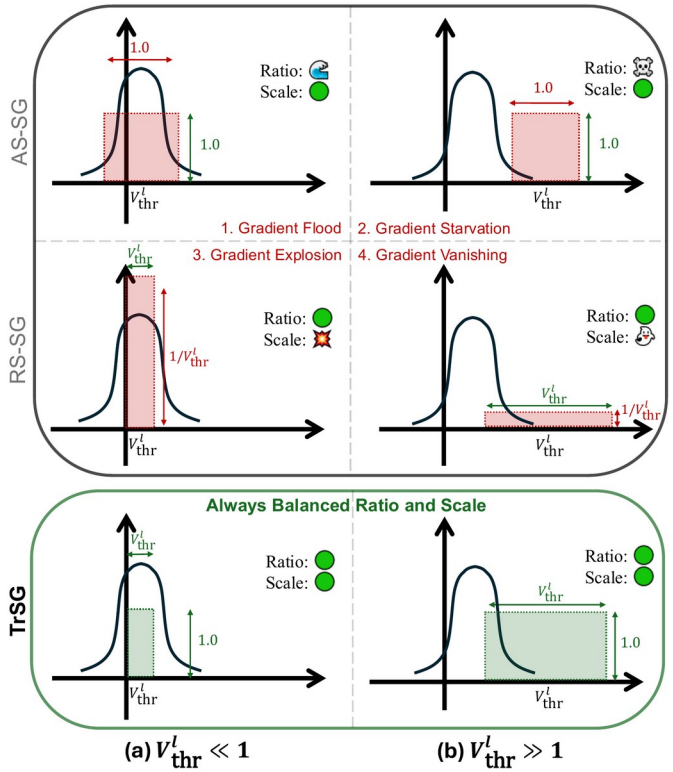
Gradient-window vs membrane distribution



Same when V_thr is fixed. Once V_thr trains, AS-SG and RS-SG break in opposite ways.

Four Failure Modes — and One Fix

Both window width AND gradient magnitude must scale correctly with V_thr



Each colored box = active gradient region. width = window · height = magnitude

✗ Flood	✗ Starvation	✗ Explosion	✗ Vanishing	✓ Balanced
AS-SG · V_thr ≪ 1	AS-SG · V_thr ≫ 1	RS-SG · V_thr ≪ 1	RS-SG · V_thr ≫ 1	TrSG · all regimes

TrSG — One-Line Fix

Multiply the threshold on the forward path; chain rule cancels the bad factor

The recipe

1 **Relative-scale window**

$$x = M / V\_thr - 1$$

2 **Multiply on forward**

$$O[t] = V\_thr \cdot S[t]$$

3 **Chain rule cancels**

$$\partial O / \partial M = f'(x)$$

Chain-rule cancellation

Window adapts with V_thr · gradient magnitude is threshold-invariant.

Inference Stays Binary

TrSG is a training-only trick — neuromorphic hardware never sees it

Training

Forward: $O[t] = V_{thr} \cdot S[t]$ (real-valued — gradient flows cleanly through V_{thr})

At deployment, absorb V_{thr} into the next layer's weights

$W'_{l \to l+1} = V_{thr}^{l} \cdot W_{l \to l+1}$ (one-time rescaling per layer)

Forward at test time: $O[t] = S[t] \in \{0, 1\}$ (standard LIF — back to binary spikes)

Standard LIF operators at inference. Hardware compatibility preserved.

Why TrSG Works — A Stable Gradient

Three stability metrics tell the same story across extreme V_thr

AbsStr

mean |gradient|

low → vanishing, high → exploding

RatioAG

% neurons with active gradient

low → starvation, high → flooding

GradCV

std / mean of |gradient|

high → unstable, hard-to-tune updates

Stress-test metrics at epoch 100 — CIFAR-100, ResNet-19, τ=2.0 frozen

V_thr = 0.1 (small threshold)

SG	AbsStr	RatioAG	GradCV
AS-SG	0.0145	90.20%	1.22
RS-SG	NaN	0.00%	NaN
TrSG	0.0153	21.45%	0.87

V_thr = 2.0 (large threshold)

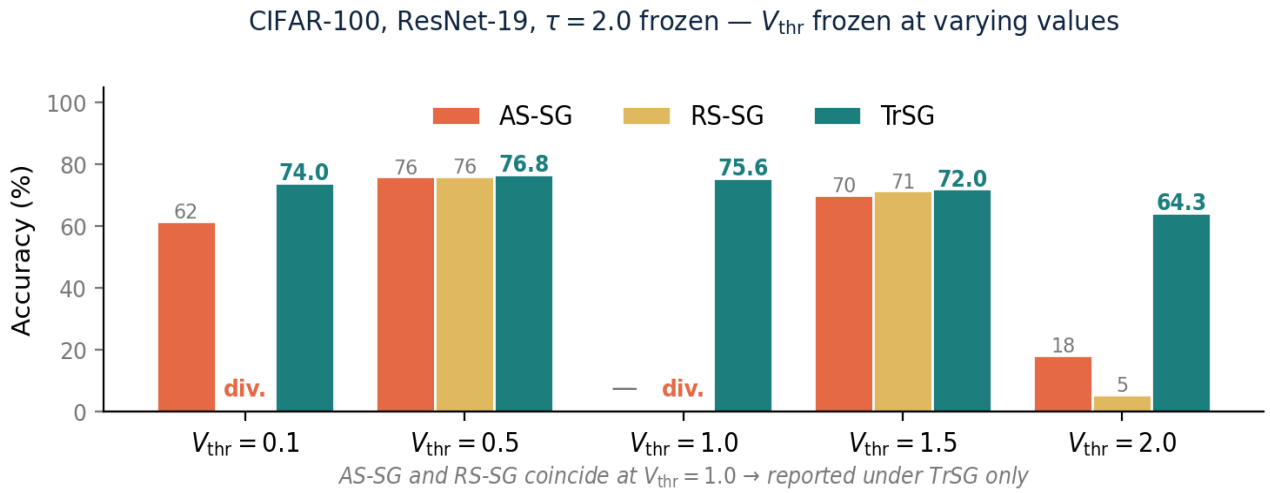
SG	AbsStr	RatioAG	GradCV
AS-SG	0.0000	0.00%	0.000
RS-SG	0.0021	0.10%	2.25
TrSG	0.0323	6.90%	0.73

TrSG keeps all three metrics in the healthy range — at every threshold.

TrSG Robust Across All V_thr

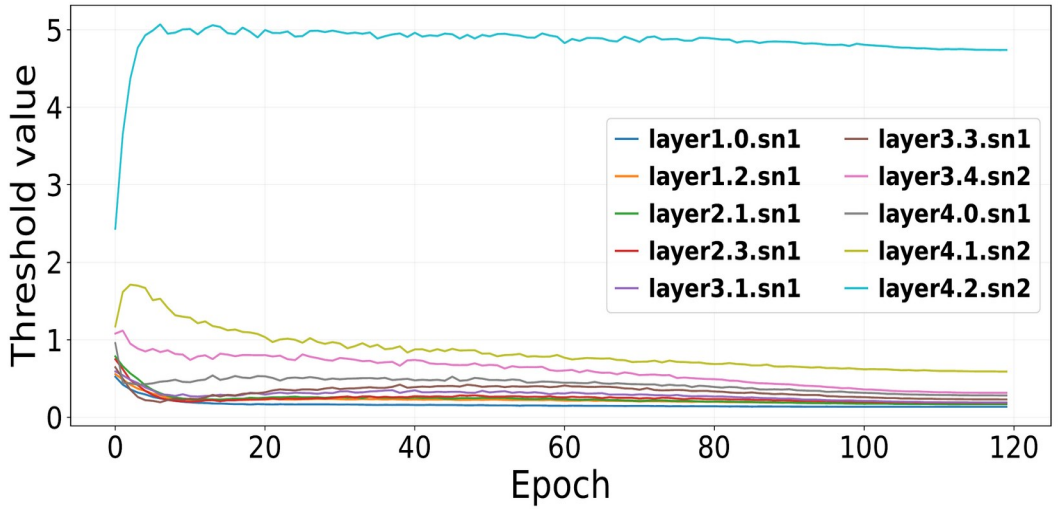
Direct accuracy evidence + real trainings actually visit the extreme regimes

Accuracy with V_thr frozen at varying values



CIFAR-100, ResNet-19, τ=2.0 frozen

Trainable V_thr trajectories — ImageNet



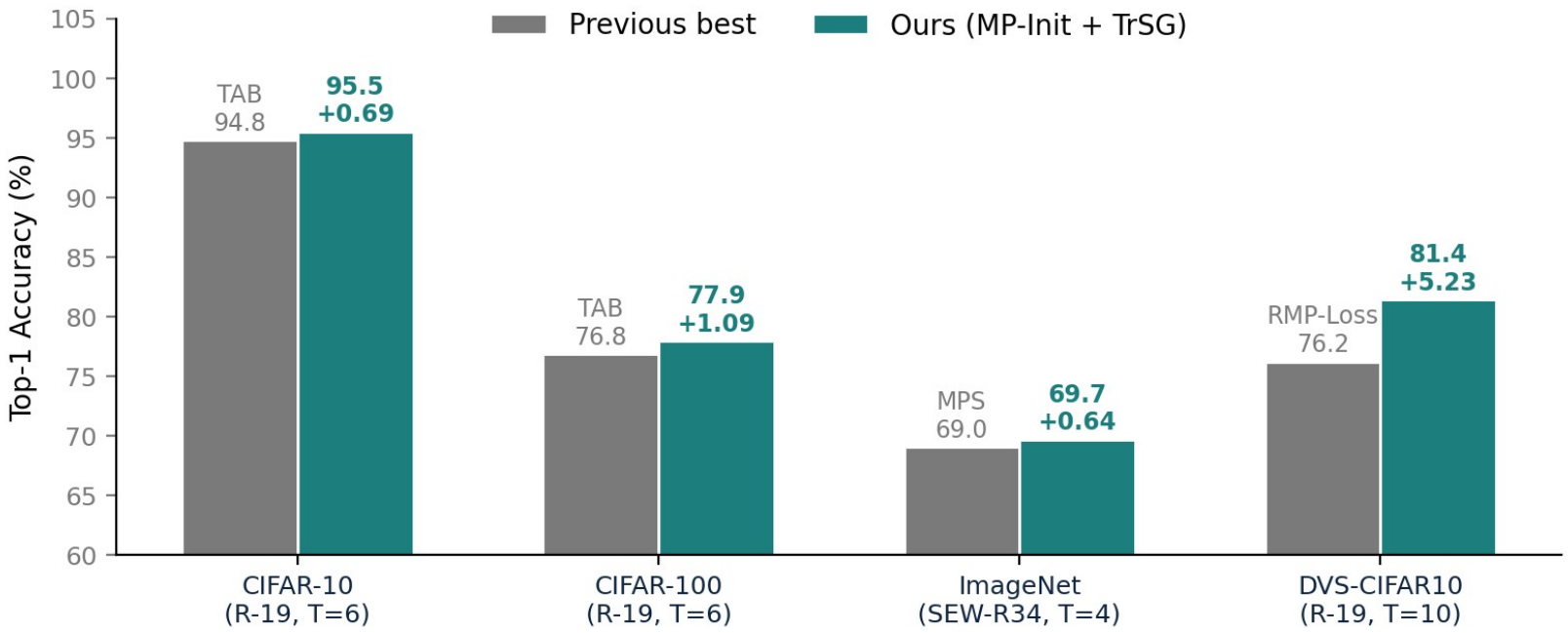
ResNet-34 on ImageNet · 120 epochs

- Frozen V_thr accuracy: TrSG wins at every threshold; AS-SG/RS-SG fail outside V_thr ≈ 1
- Real trainings drift V_thr ≪ 1 in early layers and ≫ 1 in deep layers — exactly the failure regimes

Threshold-robustness is a practical necessity, not a theoretical edge case.

Main Results — SOTA Across the Board

Standard LIF neurons · no AutoAugment / Cutout · three trials



+1.09 %pt

CIFAR-100 · beats TAB

+0.64 %pt

ImageNet · beats MPS

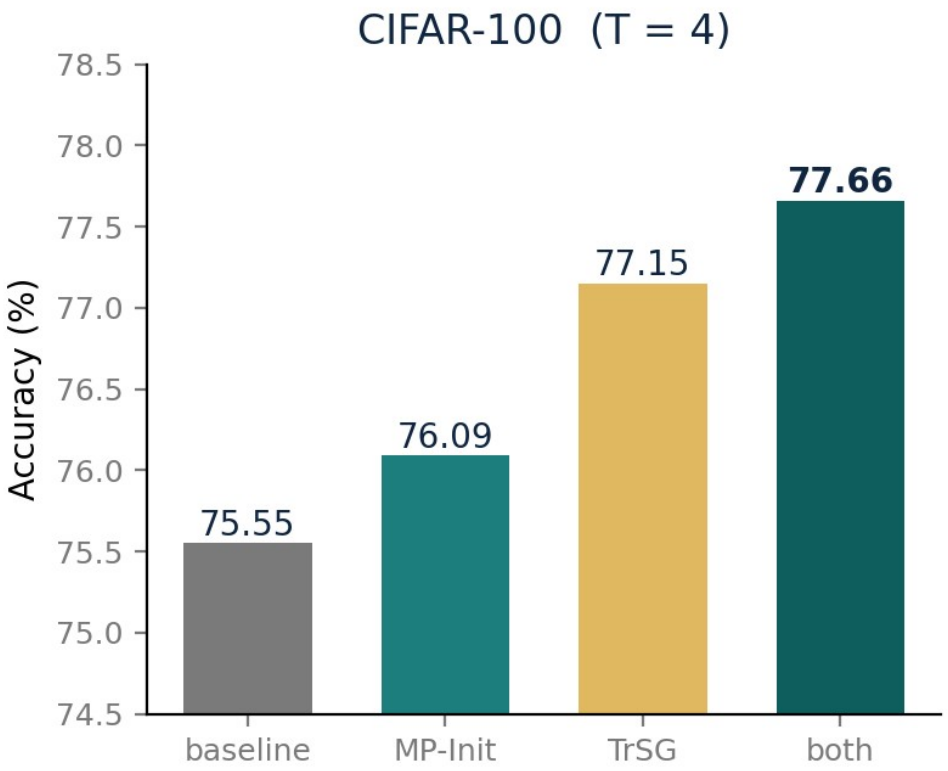
+5.23 %pt

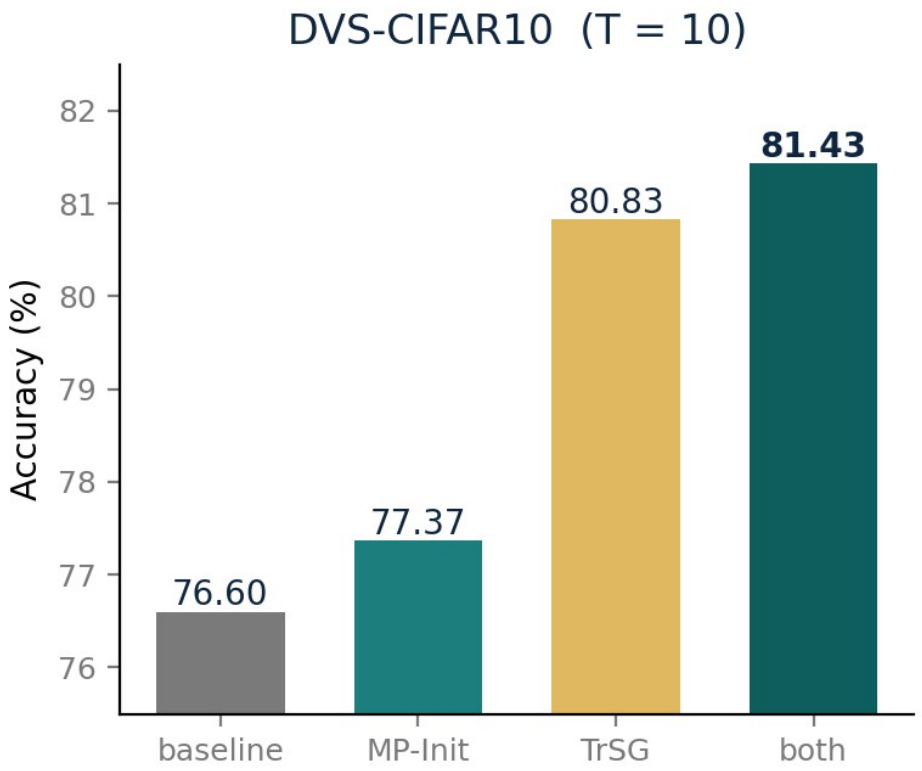
DVS-CIFAR10 · beats RMP-Loss

Biggest gain on the most challenging (event-based) dataset — DVS-CIFAR10 +5.23 %pt

Ablation — Independent and Complementary

Each method helps alone; gains stack additively (largest combined effect on event-based data)





Takeaway

MP-Init alone

+0.5 ~ 0.8

TrSG alone

+1.6 ~ 4.2

Both

+2.1 ~ 4.8

The two fixes are orthogonal — diagnoses aim at different components, not the same problem twice.

Beyond Classification

Transformer SNNs · object detection · energy preserved

Transformer SNN (QKFormer, ImageNet, T=4)

Baseline 78.80 → + ours 79.90 (+1.10 %pt)
AS-SG 77.09 → RS-SG 78.70 → TrSG 79.90
Drop-in SG swap on a state-of-the-art Transformer SNN.

Object detection (COCO-2017, EMS-ResNet10, T=3)

mAP@0.5 0.291 → 0.305 (+0.014)
mAP@0.5:0.95 0.138 → 0.147 (+0.009)
First evidence the principles transfer beyond classification.

Energy under matched configuration (CIFAR-100)

Baseline 1.57 mJ · 75.58 %
+ MP-Init + TrSG 1.67 mJ · 77.68 %
+2.10 %pt at +6 % energy → orders of magnitude better than spike-rate-multiplying baselines (cf. Pareto frontier on slide 9).

All four surrogate shapes

Rectangular · Triangular · Arctan · Sigmoid
TrSG remains the best across initial V_{thr}, initial τ, and surrogate shape — never tied for second.

No architectural changes, no extra losses, no per-timestep parameters — just drop in.

CONCLUSION

What This Talk Was About

Two long-standing instabilities of direct SNN training, traced and fixed.

Both fixes are minimal, principled, and preserve standard inference.

MP-Init

TCS originates in the membrane potential — fix the initial state.

Per-layer running mean of U[T].

Zero overhead, standard LIF inference.

TrSG

Threshold-induced gradient pathology — fix it with one forward-path multiplication.

Window adapts, magnitude is invariant.

Standard LIF after training.

Validated

SOTA on CIFAR-10/100, ImageNet, DVS-CIFAR10.

Generalizes to Transformer SNNs (QKFormer +1.10) and detection (COCO +0.014 mAP).

All without architectural changes.

CONCLUSION

Two instabilities, two principled fixes, one cohesive story.

LIMITATIONS

- Per-layer constant approximates π only at the mean — richer parameterizations are open.
- Hardware with fixed neuron parameters needs quantization-aware adaptation.
- i.i.d. assumption empirically supported, but tighter bounds welcome.

FUTURE WORK

- Hardware-aware adaptation: Loihi, Akida, TrueNorth
- Beyond LIF: adaptive LIF, Izhikevich, foundation-scale SNNs

Thank you. Questions?

BACKUP MATERIAL

Appendix

Q&A reference: hardware, datasets, prior work

APPENDIX · A1

Can SNNs Actually Run on Chips?

Production-grade neuromorphic silicon already exists.

Yes — and at scale.

Several research and commercial chips natively run spiking models. Our methods (MP-Init, TrSG) are training-time techniques that produce the same LIF model these chips already accept — so the trained network ports directly without architectural changes.

Chip	Year	Process	Capacity	Headline metric
IBM TrueNorth	2014	65 nm	1 M neur. 256 M syn.	70 mW · 26 pJ/event
Intel Loihi	2018	14 nm	130 K neur. 130 M syn.	23.6 pJ / op
Intel Loihi 2	2021	Intel 4	1 M neur. graded spikes	improved · multi-bit
SpiNNaker 2	2024	22 nm	152 K neur. 152 M syn. / chip	ARM-based · digital
Tianjic	2019	28 nm	40 K neur. SNN+ANN hybrid	Tsinghua · hybrid arch.
Speck (SynSense)	2022	65 nm	328 K neur. DVS on-chip	< 10 mW · always-on
BrainScaleS-2	2022	65 nm	512 analog neur.	1000× biological speed

Key takeaway

Hardware is here. The bottleneck is training quality at low T.

Refs: Merolla 2014 · Davies 2018 · Mayr 2019 · Pei 2019 · SynSense 2022 · BrainScaleS team 2022.

Where SNN Efficiency Comes From

Three structural advantages — none requires a new training scheme.

① Event-driven

No spike → no compute. Inactive neurons skip MAC and memory access entirely.

`Typical activity:  5–20 % per timestep`

② Multiplication-free

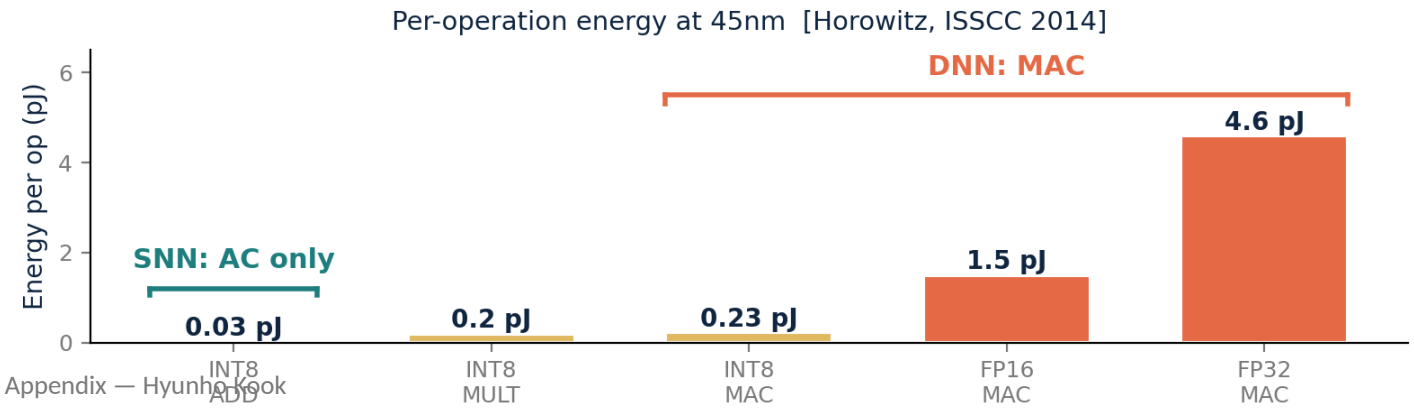
Spike ∈ {0, 1} → W·spike degenerates to W or 0. Each synapse becomes an Accumulate (AC), not a MAC.

`INT8 AC ≈ 8× cheaper than INT8 MAC (45 nm)`

③ Co-located memory

Neuromorphic cores keep synaptic weights in local SRAM, on-chip mesh routing — no DRAM trip.

`DRAM 64-bit ≈ 1300 pJ vs SRAM ≈ 5 pJ`

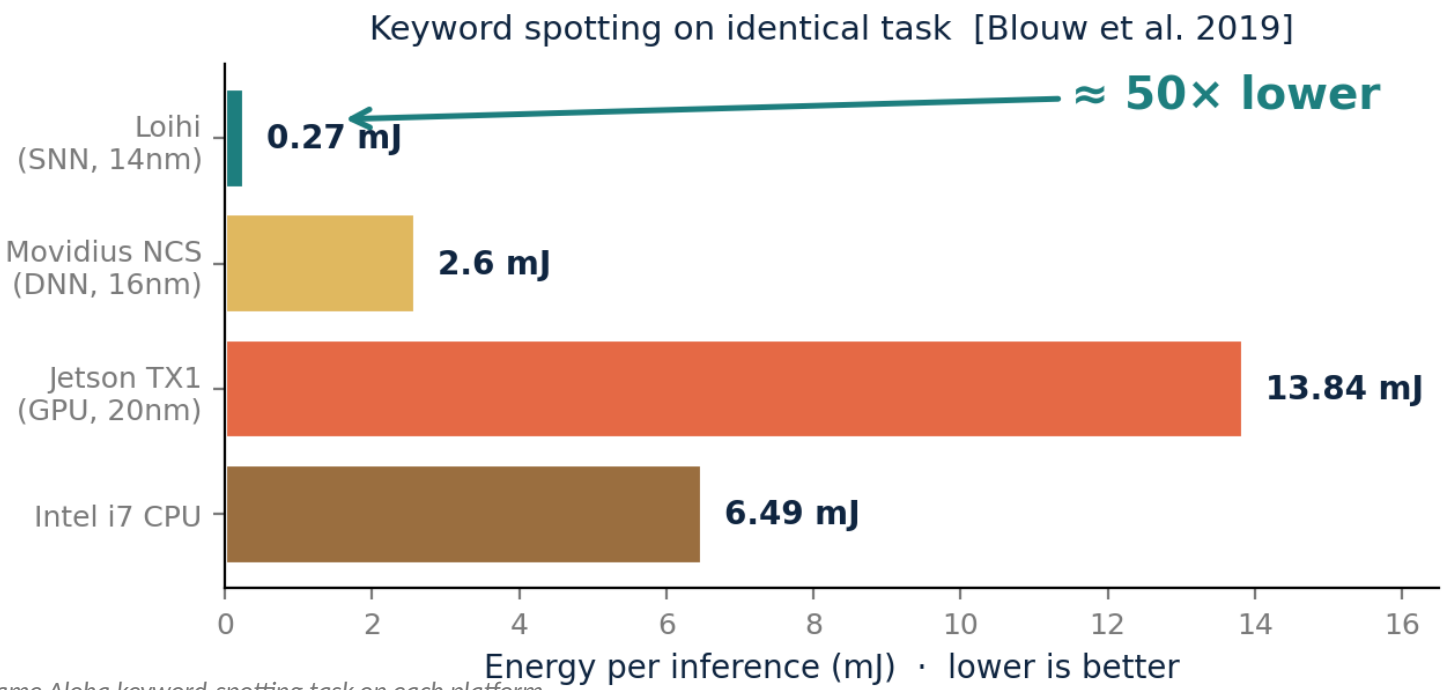


Bottom line

Efficiency is structural — comes from sparsity + binary spikes + co-located memory.
MP-Init / TrSG protect (1) by stabilizing low-T training.

Energy: SNN Chip vs DNN Platforms

Same task, four hardware platforms — what does the gap actually look like?



Same Aloha keyword-spotting task on each platform.

Sources: Blouw et al. 2019 (keyword-spotting benchmark) ·
Davies et al. 2021 (LASSO 48× vs CPU) · Horowitz ISSCC 2014 (per-op energy).

How big is the gap?

Sparse / event tasks

~50–100× vs embedded GPU
(keyword spot, gesture, sparse coding)

Static images (CIFAR / ImageNet)

~2–5× vs GPU at iso-accuracy
(per published SNN/CNN comparisons)

Why so task-dependent?

Energy ∝ T × spike-rate × neurons.
Low-T + high-sparsity → big gap.

Why this matters for our work

MP-Init enables T = 2 SOTA → directly multiplies energy savings.

What Is an Event-Based Dataset?

Output of asynchronous DVS cameras — sparse, fast, high dynamic range.

DVS: Dynamic Vision Sensor

Each pixel asynchronously emits an event when its log-intensity changes by a threshold.
Output: a stream of (x, y, t, polarity) tuples.

µs	>120 dB	Sparse	No blur
temporal resolution	dynamic range	only changes emit	no exposure window

In this thesis

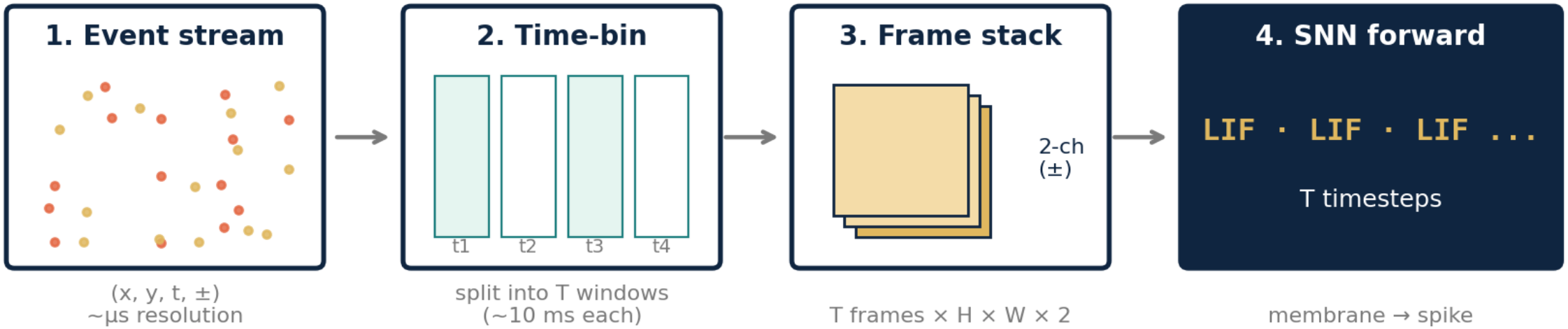
DVS-CIFAR10 (event-based image classification, T = 10).

Dataset	Description	Size
DVS-CIFAR10	CIFAR-10 on monitor + saccading DVS	10 K
N-MNIST	MNIST shown to event camera	70 K
N-Caltech101	Caltech101 in event form	9.1 K
DVS-Gesture (IBM)	11 hand gestures, 29 subjects	1.3 K clips
SHD	Spiking Heidelberg Digits (audio events)	10 K
Gen1 / 1Mpx	Prophesee automotive (object detect.)	39 h / 14 h
ASL-DVS	American Sign Language (event)	100 K

Highlighted: dataset used in this thesis.

From DVS Events to SNN Input

Asynchronous events become T frames that the LIF stack consumes.



Why this matters for MP-Init

The first frame after binning has no history → the membrane starts at zero, far from π. That's exactly the gap MP-Init closes.

How Were SNNs Trained Before?

From biological learning rules to modern surrogate-gradient BPTT.

Spike-Timing-Dependent Plasticity

Biological Hebbian rule.
Unsupervised, local; works on small nets only.

Bi & Poo 1998
Diehl & Cook 2015

Train ANN → convert weights

Pretrained ReLU CNN → spike rates.
Early methods: T = 100–2000.
Modern (QCFS '23, Hao '23): T ≤ 8.
Still cannot consume native event data.

Cao 2015 · Rueckauer 2017
Bu 2023 (QCFS) · Hao 2023

Surrogate gradient (STBP / BPTT)

Replace Heaviside backward with a smooth function.
Few timesteps; works on event data; modern dominant.

Bohte 2002 · Wu 2018
Neftci 2019

Learnable LIF parameters

Train V_thr, τ end-to-end → adaptive neurons.
This thesis: stabilize that training pathway.

Fang 2021 · Wang 2022
Rathi 2023 · ours

Where Prior Work Falls Short

TCS methods miss the membrane potential; threshold-learning methods don't diagnose the gradient.

TCS / Normalization in SNNs

Method	What it does	What it misses
BNTT '21	BN on synaptic input current	*U[t] drift not corrected*
tdBN '20	Threshold-dependent BN on activations	*Pre-spike BN; U[t] persists*
TEBN '22	Timestep-specific affine on outputs	*Membrane drift untouched*
TAB '24	Temporal-adaptive output normalization	*Membrane TCS persists*

Our take (MP-Init)

Fix the cause, not symptom — align M[t] with stationary π directly.

Trainable LIF parameters (V_thr, τ)

Method	What's learned	What it misses
PLIF '21	Leakage τ only	*V_thr fixed*
LTMD '22	V_thr + τ (heavy clipping)	*Hides instability*
DIET-SNN '23	V_thr + τ (gradient clip)	*Doesn't classify failure*
Ours (TrSG)	**V_thr + τ, scale-invariant**	—

Our take (TrSG)

Diagnose first (4 failure modes), then cancel the 1/V_thr factor by construction.